CHAPTER 13

Optical Absorption and Recombination of Bulk Semiconductors

- Introduction to quantum theory
- Optical absorption of semiconductors
- Exciton-enhancement of optical absorption
- Optical recombination

Semiconductor Materials

An Introduction to Basic Principles

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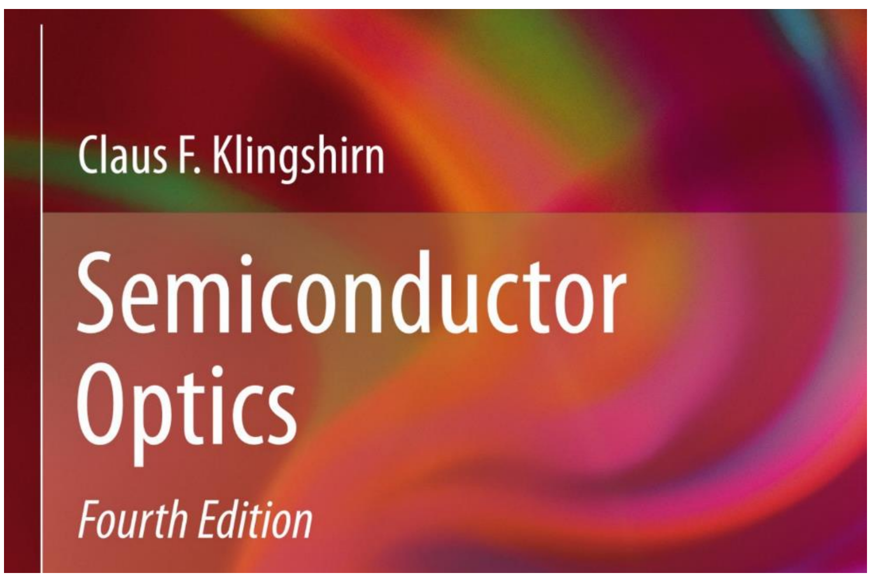
The Physics of Semiconductors

An Introduction Including Devices and Nanophysics



13.1 Introduction to quantum theory

If a quantum-mechanical system, such as an atom, is known to be in a particular quantum-mechanical state, which we designate s, we can describe all of the physical properties of the system in terms of the wavefunction $\Psi_s(\mathbf{r},t)$, which obeys Schrödinger's equation

$$i\hbar \frac{\partial \psi_s(r,\mathrm{t})}{\partial t} = \hat{H}\psi_s(r,\mathrm{t}), \tag{13.1.1a}$$

$$\hat{H} = \hat{H}_0 + \hat{H}', \tag{13.1.1b}$$

Where $\hat{H}_0$ is the Hamiltonian of a free atom and $\hat{H}'$ is the interaction with the external field. For the case in which no external field is applied to the atom, the Hamiltonian is simply equal to $\hat{H}_0$ and Schrödinger equation possesses solution in the form of energy eigenstates. Theses states are also known as stationary states, because the time evolution of these states is given by a simple exponential phase factor as follows:

$$\psi_{\mathrm{n}}(r,\mathrm{t}) = u_{\mathrm{n}}(r)\exp(-\mathrm{i}\omega\mathrm{t}) \tag{13.1.2}$$

Time-independent Schrödinger's equation

By substituting this form of wavefunction into the Schrödinger's equation (13.1.1a), we find that the spatially varying part of the wavefunciton $u_n(\boldsymbol{r})$ must satisfy the eigenvalue equation, known as time-independent Schrödinger's equation,

$$\hat{H}_0 u_n(\boldsymbol{r}) = E_n u_n(\boldsymbol{r}), \tag{13.1.3}$$

Where $E_n = \hbar\omega_n$, $u_n(\boldsymbol{r})$ are called the energy eigensolutions of Eq. (13.1.3) and are assumed to be orthonormal in that they obey the relation

$$\int u_m^*(\boldsymbol{r}) u_n(\boldsymbol{r}) d^3r = \delta_{mn}, \tag{13.1.4}$$

The wavefunction of state s, $\Psi_s(\mathbf{r},t)$, can be expressed in terms of eigenstates $u_n(\boldsymbol{r})$

$$\psi_s(\boldsymbol{r},t) = \sum_n C_n^s(t) u_n(\boldsymbol{r}), \tag{13.1.5}$$

The expansion coefficient gives the probability that the atom is in the energy eigenstate *n* at time *t*.

Introducing Eq. (13.1.5) into Schrödinger's equation (13.1.1a), we have

$$i\hbar \sum_{n} \frac{dC_n^s(t)}{dt} u_n(r) = \sum_{n} C_n^s(t) \hat{H}\, u_n(r), \qquad (13.1.6)$$

Multiplying each side from the left by $u_m^*(\boldsymbol{r})$, integrate over all space, and utilize Eq. (13.1.4), we have

$$i\hbar \frac{dC_m^s(t)}{dt} = \sum_{n} H_{mn} C_n^s(t), \qquad (13.1.7)$$

Where

$$H_{mn} = \int u_m^*(r) \hat{H} u_n(r)\, d^3r \qquad (13.1.8)$$

The expectation value of any observable quantity can be calculated in terms of wavefunction of the system.

$$\langle A \rangle = \int \Psi_s^* \hat{A} \Psi_s d^3 r \qquad (13.1.9)$$

This relationship can be conveniently written in Dirac notation

$$\langle A \rangle = \langle \psi_s | \hat{A} | \psi_s \rangle = \langle s | \hat{A} | s \rangle. \tag{13.1.10}$$

The expectation value <A> can also be expressed in terms of the probability amplitude. By introducing Eq. (13.1.5) into Eq. (13.1.9), we obtain

$$\langle A \rangle = \sum_{mn} C_m^{s*} C_n^s A_{mn}, \tag{13.1.11}$$

Where

$$A_{mn} = \langle u_m | \hat{A} | u_n \rangle = \int u_m^* \hat{A} u_n d^3 r. \tag{13.1.12}$$

Fermi golden rule

Assume the Hamiltonian of the system is H_0, and is at initial state $|i\rangle$, with a time-dependent perturbation H′, the atom will transit to state $|f\rangle$. The transition probability per time is given by

$$w_{\mathrm{fi}}(\hbar\omega) = \frac{2\pi}{\hbar}\,|H'_{\mathrm{fi}}|^2\,\delta(E_{\mathrm{f}} - E_{\mathrm{i}} - \hbar\omega)\ , \qquad (13.1.13a)$$

where $\hbar\omega$ is the photon energy, E_i (E_f) is the energy of the initial (final) state.

$$H'_{fi} = \langle f|H'|i\rangle \qquad (13.1.13b)$$

13.2 Optical absorption of semiconductors

13.2.1 Optical absorption of semiconductors – overview

In general, there are several optical absorption processes, and each of these contributes to the total absorption. These processes include:

- Phonon absorption
- Plasmon absorption
- Fundamental absorption
- Exciton absorption
- Absorption due to dopants and imperfections
- Absorption due to intraband transitions
- Free carrier absorption

Spectral regions of optical absorption

Table 9.1. Spectral ranges with relevance to semiconductor optical properties

Range		wavelengths	energies
deep ultraviolet	DUV	< 250 nm	> 5 eV
ultraviolet	UV	250–400 nm	3–5 eV
visible	VIS	400–800 nm	1.6–3 eV
near infrared	NIR	800 nm–2 μm	0.6–1.6 eV
mid-infrared	MIR	2–20 μm	60 meV–0.6 eV
far infrared	FIR	20–80 μm	1.6–60 meV
THz region	THz	> 80 μm	< 1.6 meV

$$E = \frac{1240\,\mathrm{eV\,nm}}{\lambda}$$

Optical absorptions of semiconductors

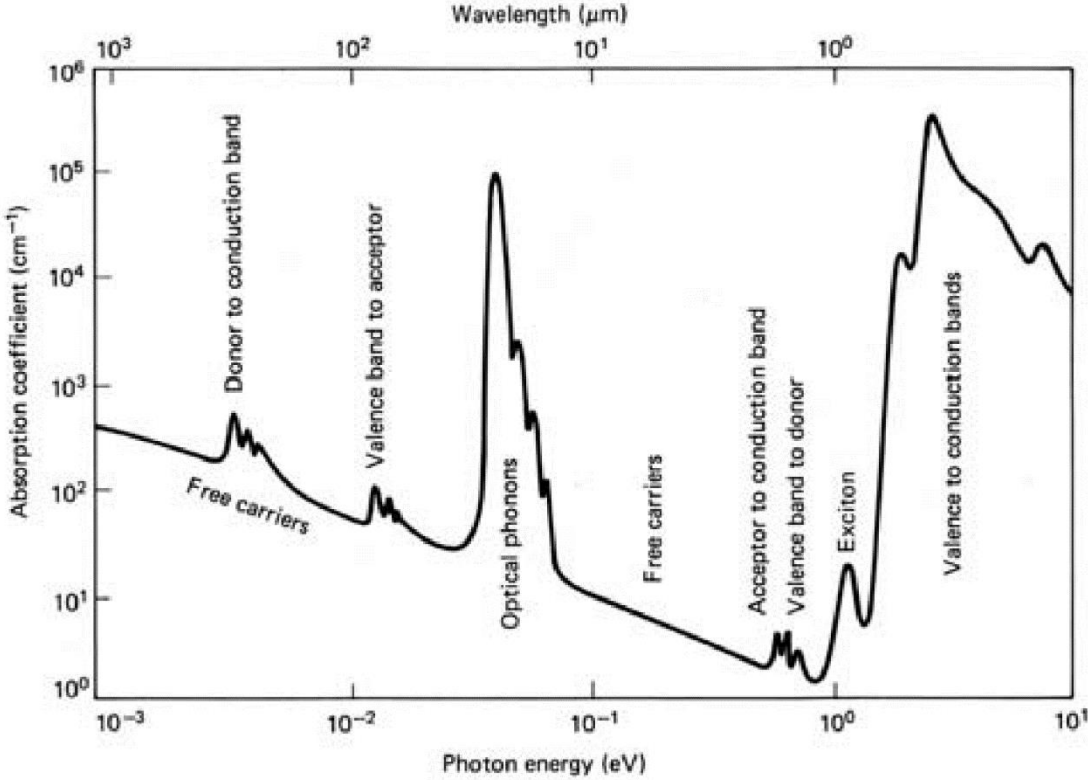


Fig. 9.1. Schematic absorption spectrum of a typical semiconductor. With permission from [278], ©1989 Prentice-Hall

13.2.2 Fundamental absorption of semiconductors - Band-band absorption

- When the minimum of the conduction band and the maximum of the valence band occur at the same value of the wave vector k, transitions are direct, and the semiconductor is referred to as a direct-gap semiconductor.
- If the band extrema do not occur at the same wave vector k, transitions are indirect, and the semiconductor is referred to as an indirect-gap semiconductor.

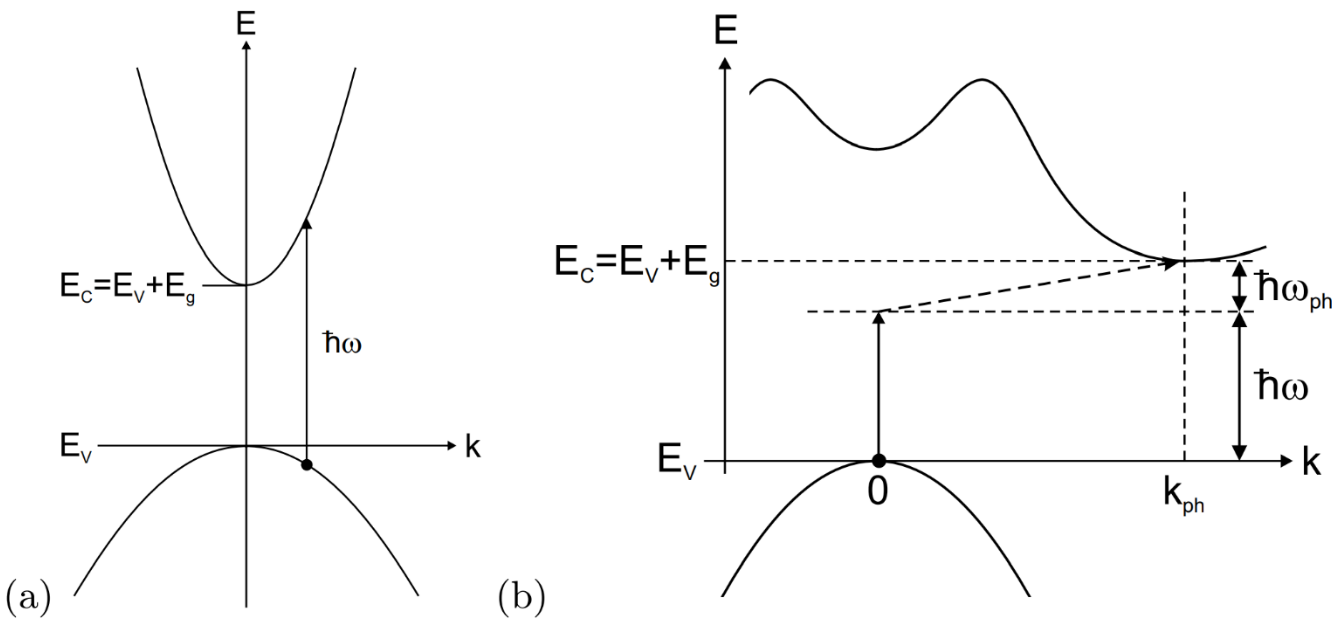
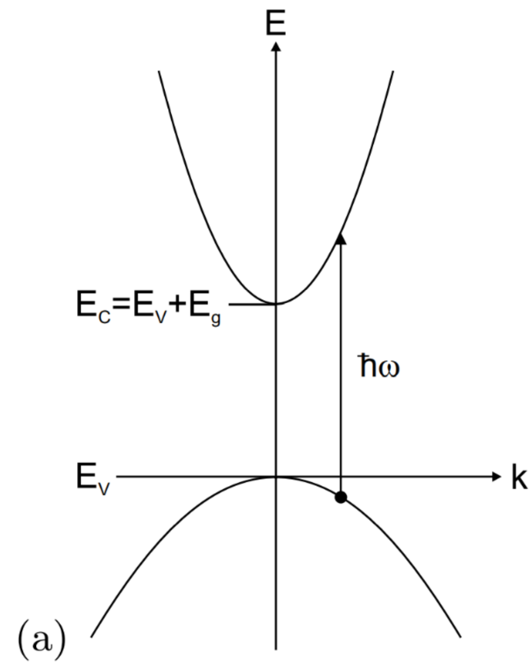


Fig. 9.6. (**a**) Direct optical transition and (**b**) indirect optical transitions between valence and conduction bands. The photon energy is $\hbar\omega$. The indirect transition involves a phonon with energy $\hbar\omega_{\mathrm{ph}}$ and wavevector k_{ph}.

(a) Direct transition

- In the fundamental absorption process, a photon of energy greater than the bandgap energy E_g excites an electron from the valence band to the conduction band.
- The minimum of the conduction band and the maximum of the valence band occur at the same value of the wave vector k.
- Both energy and momentum must be conserved in this process. Since the photon momentum is small, the absorption process should essentially conserve the electron momentum



$$\frac{(\hbar\mathbf{k}_i)^2}{2m_h} + \hbar\omega = Eg + \frac{(\hbar\mathbf{k}_f)^2}{2m_e} \qquad (13.2.1)$$

$$\mathbf{k}_i \approx \mathbf{k}_f \qquad (13.2.2)$$

Optical absorption of direct bandgap semiconductors

$$R(\hbar\omega) = \frac{2\pi}{\hbar}\int_{\mathbf{k}_c}\int_{\mathbf{k}_v} |\langle c|\mathbf{H}_{\mathrm{em}}|v\rangle|^2\,\delta\left(E_{\mathrm{c}}(\mathbf{k}_{\mathrm{c}}) - E_{\mathrm{v}}(\mathbf{k}_{\mathrm{v}}) - \hbar\omega\right)\mathrm{d}\mathbf{k}_{\mathrm{c}}\mathrm{d}\mathbf{k}_{\mathrm{v}} \tag{13.2.3a}$$

$$\mathbf{H}_{\mathrm{em}} = -\frac{q}{m}\mathbf{A}\cdot\mathbf{p} = \frac{\mathrm{i}q\hbar}{m}\mathbf{A}\cdot\nabla \approx q\mathbf{r}\cdot\mathbf{E}\,. \tag{13.2.3b}$$

$$|\langle c|\mathbf{H}_{\mathrm{em}}|v\rangle|^2 = \frac{e^2\,|A|^2}{m^2}\,|\langle c|\hat{\mathbf{e}}\cdot\mathbf{p}|v\rangle|^2 \tag{13.2.4a}$$

$$\langle c\,|\hat{\mathbf{e}}\cdot\mathbf{p}|v\rangle|^2 = \frac{1}{3}|\mathbf{p}_{\mathrm{cv}}|^2 = M_{\mathrm{b}}^2 \tag{13.2.4b}$$

$$R(\hbar\omega) = \frac{2\pi}{\hbar}\left(\frac{e}{m\omega}\right)^2\left|\frac{\mathcal{E}(\omega)}{2}\right|^2|\mathbf{p}_{\mathrm{cv}}|^2\int_{\mathbf{k}}\delta\left(E_{\mathrm{c}}(\mathbf{k}) - E_{\mathrm{v}}(\mathbf{k}) - \hbar\omega\right)\mathrm{d}\mathbf{k} \tag{13.2.5}$$

$$\epsilon_{\mathrm{i}} = \frac{1}{4\pi\epsilon_0}\left(\frac{2\pi e}{m\omega}\right)^2|\mathbf{p}_{\mathrm{cv}}|^2\int_{\mathbf{k}}\delta\left(E_{\mathrm{c}}(\mathbf{k}) - E_{\mathrm{v}}(\mathbf{k}) - \hbar\omega\right)\mathrm{d}\mathbf{k} \tag{13.2.6a}$$

$$\epsilon_{\mathrm{r}} = 1 + \int_{\mathbf{k}}\frac{e^2}{\epsilon_0 m\omega_{\mathrm{cv}}^2}\frac{2|\mathbf{p}_{\mathrm{cv}}|^2}{m\hbar\omega_{\mathrm{cv}}}\frac{1}{1-\omega^2/\omega_{\mathrm{cv}}^2}\mathrm{d}\mathbf{k}\,, \tag{13.2.6b}$$

$$f = \frac{e^2}{\epsilon_0 m\omega_{\mathrm{cv}}^2}\frac{2|\mathbf{p}_{\mathrm{cv}}|^2}{m\hbar\omega_{\mathrm{cv}}} \tag{13.2.7}$$

Joint Density of States (JDOS): it quantifies the number of electron-hole pairs available for interband optical transitions between the valence band (VB) and conduction band (CB) of a semiconductor.

Absorption coefficient

$$\alpha(\hbar\omega) = A \sum W_{\mathrm{if}}^{\mathrm{ab}} n_{\mathrm{i}} n'_{\mathrm{f}} \tag{13.2.8}$$

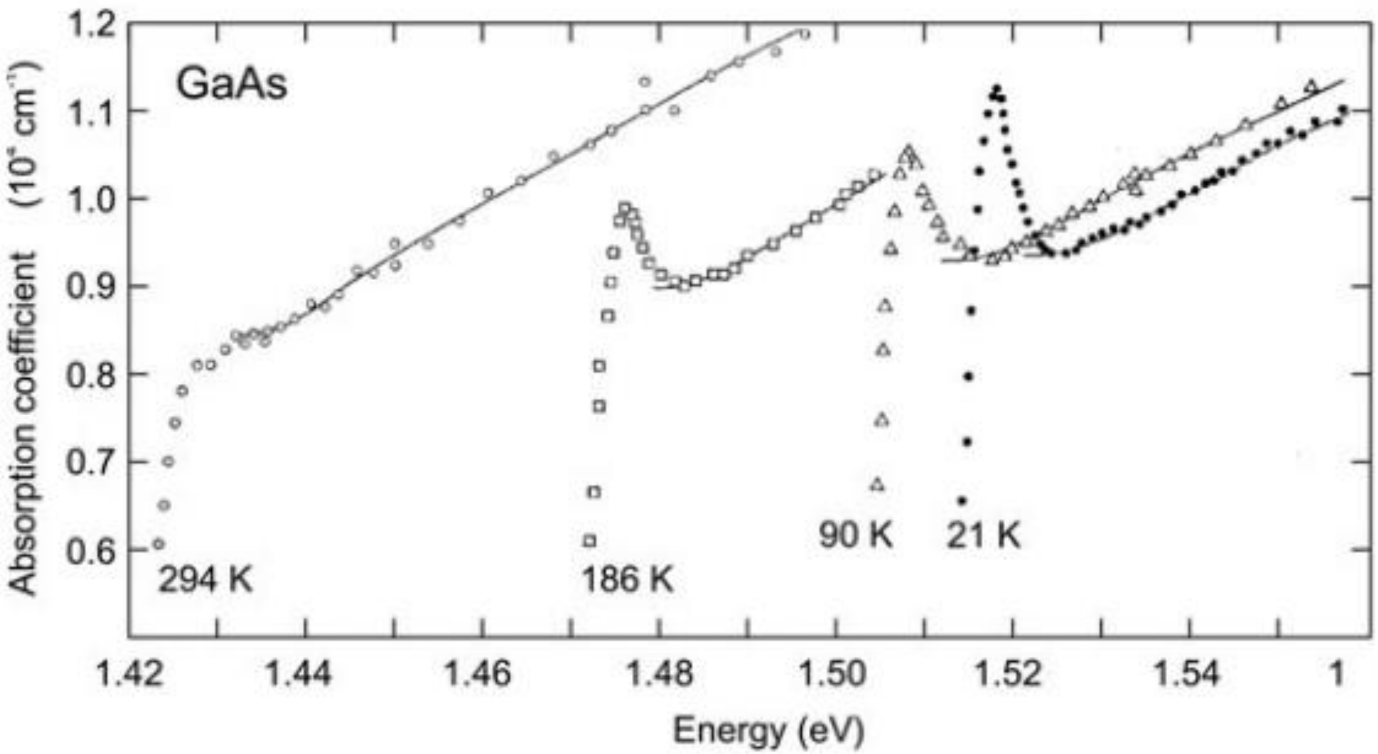
$$\alpha_{\mathrm{d}}(\hbar\omega) = \begin{cases} A(\hbar\omega - E_{\mathrm{g}})^{1/2}, & \hbar\omega \geqslant E_{\mathrm{g}} \\ 0, & \hbar\omega < E_{\mathrm{g}} \end{cases} \tag{13.2.9a}$$

$$A \approx \frac{(2m_{\mathrm{r}}^{*})^{3/2} e^{2}}{\eta c m_{\mathrm{e}}^{*} \hbar^{2} \varepsilon_{0}} \tag{13.2.9b}$$

$$m_{\mathrm{r}}^{*} = \frac{m_{\mathrm{h}}^{*} m_{\mathrm{e}}^{*}}{m_{\mathrm{h}}^{*} + m_{\mathrm{e}}^{*}} \tag{13.2.9c}$$

$m_e^* = m_e^* = m_0$, $\eta = 4$, $A \approx 2\text{x}10^4\ \text{cm}^{-1}$

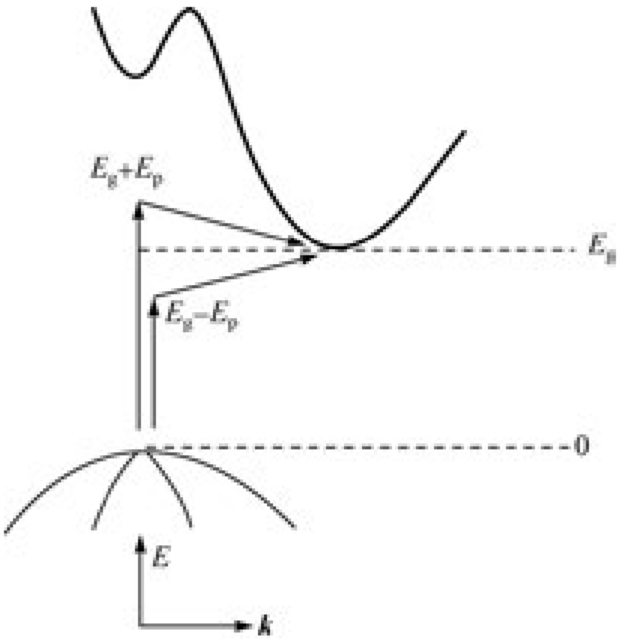
$$\alpha_{\mathrm{d}}(\hbar\omega) = \begin{cases} \approx 2 \times 10^{4} (\hbar\omega - E_{\mathrm{g}})^{1/2}, & \hbar\omega \geqslant E_{\mathrm{g}} \\ = 0, & \hbar\omega < E_{\mathrm{g}} \end{cases} \tag{13.2.10}$$



Absorption spectra of GaAs close to the bandgap at different temperatures

(b) Indirect transition

- The band extrema do not occur at the same wave vector k, the material is referred to as an indirect-gap semiconductor.
- For momentum conservation in such semiconductors, optical transitions (absorption and photoluminescence) are indirect, a participation of an extra particle, i.e., phonon, is required.
- The probability of optical transitions (absorption and luminescence) is substantially lower, compared with direct transitions. Therefore, fundamental absorption in indirect-gap semiconductors is usually weaker as compared with direct-gap materials.



Absorption with phonon emission

$$\hbar\omega_e = E_c(\boldsymbol{k}_c) - E_v(\boldsymbol{k}_v) + E_p \qquad (13.2.11a)$$

$$\hbar\omega \leqslant E_g + E_p \text{ 时 } \alpha_e(\hbar\omega) = 0 \qquad (13.2.11b)$$

Absorption with phonon absorption

$$\hbar\omega_a = E_c(\boldsymbol{k}_c) - E_v(\boldsymbol{k}_v) - E_p \qquad (13.2.12a)$$

$$\hbar\omega \leqslant E_g - E_p \text{ 时 } \alpha_a(\hbar\omega) = 0 \qquad (13.2.12b)$$

Absorption coefficient with phonon emission

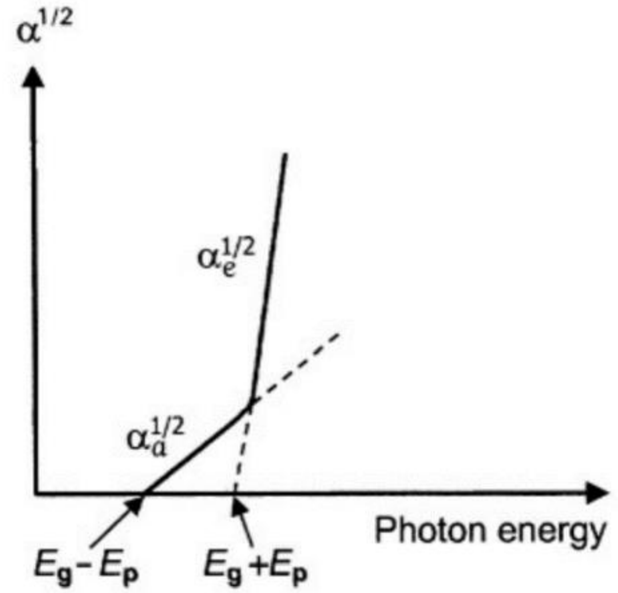
$$\alpha_e(\hbar\omega) = \begin{cases} \dfrac{B(\hbar\omega - E_g - E_p)^2}{1 - \exp(-E_p/k_B T)}, & \hbar\omega > (E_g + E_p) \\ 0, & \hbar\omega \leqslant (E_g + E_p) \end{cases} \quad (13.2.13)$$

Absorption coefficient with phonon absorption

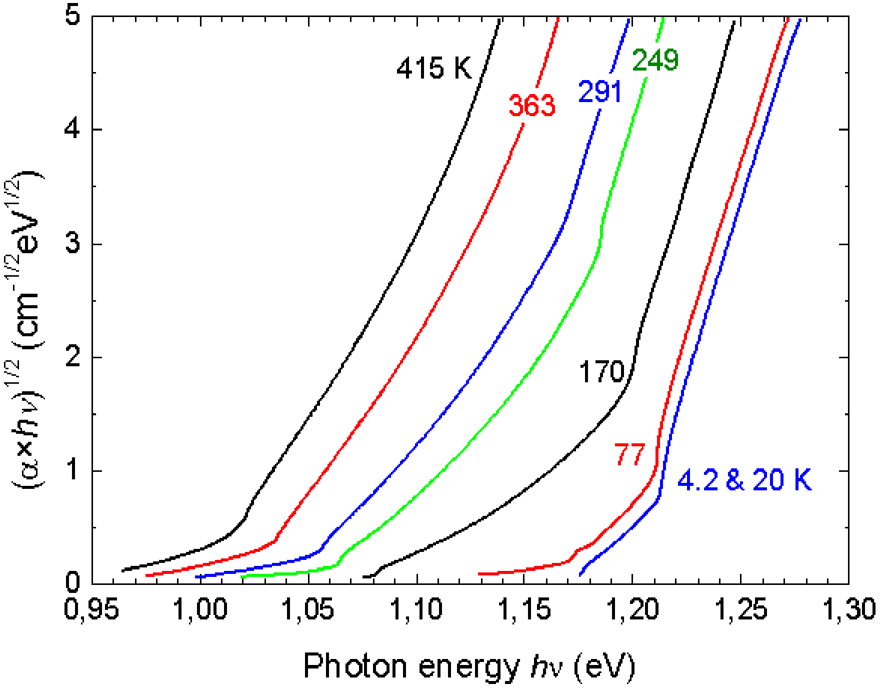
$$\alpha_a(\hbar\omega) = \begin{cases} \dfrac{B(\hbar\omega - E_g + E_p)^2}{\exp(E_p/k_B T) - 1}, & \hbar\omega > (E_g - E_p) \\ 0, & \hbar\omega \leqslant (E_g - E_p) \end{cases} \quad (13.2.14)$$

The total absorption coefficient with phonon emission and absorption

$$\alpha_i(\hbar\omega) = \begin{cases} B\left[\dfrac{(\hbar\omega - E_g - E_p)^2}{1 - \exp(-E_p/k_B T)} + \dfrac{(\hbar\omega - E_g + E_p)^2}{\exp(E_p/k_B T) - 1}\right], & \hbar\omega > (E_g + E_p) \\ B\dfrac{(\hbar\omega - E_g + E_p)^2}{\exp(E_p/k_B T) - 1}, & E_g - E_p \leqslant \hbar\omega \leqslant E_g + E_p \\ 0, & \hbar\omega \leqslant (E_g - E_p) \end{cases} \quad (13.2.15)$$

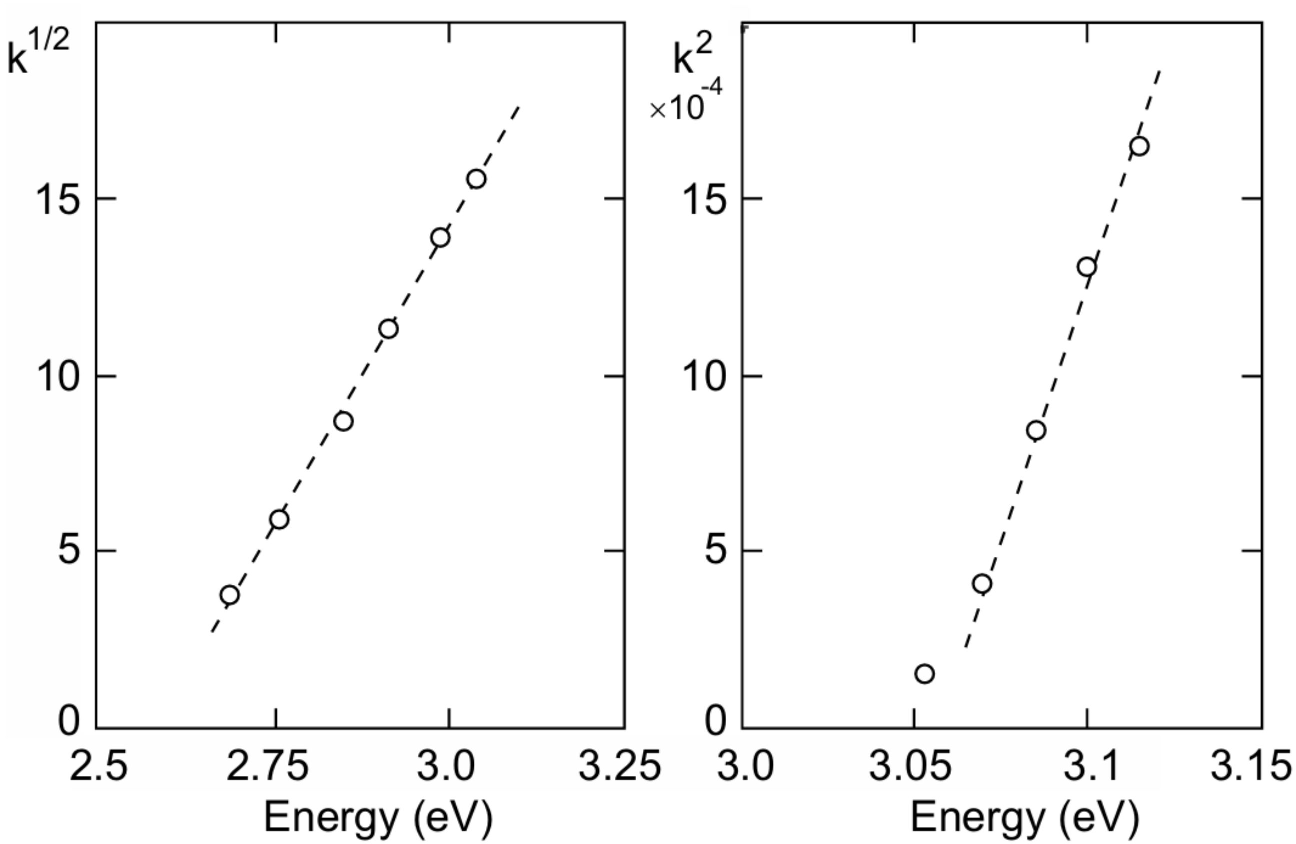


Schematic diagram of absorption spectrum of indirect semiconductors



Absorption spectrum of Si at various temperatures

Direct absorption in indirect semiconductors



Absorption of $BaTiO_3$ at room temperature. Experimental data (circles) with fits (dashed lines) $\propto E^2$ and $\propto E^{1/2}$, respectively.

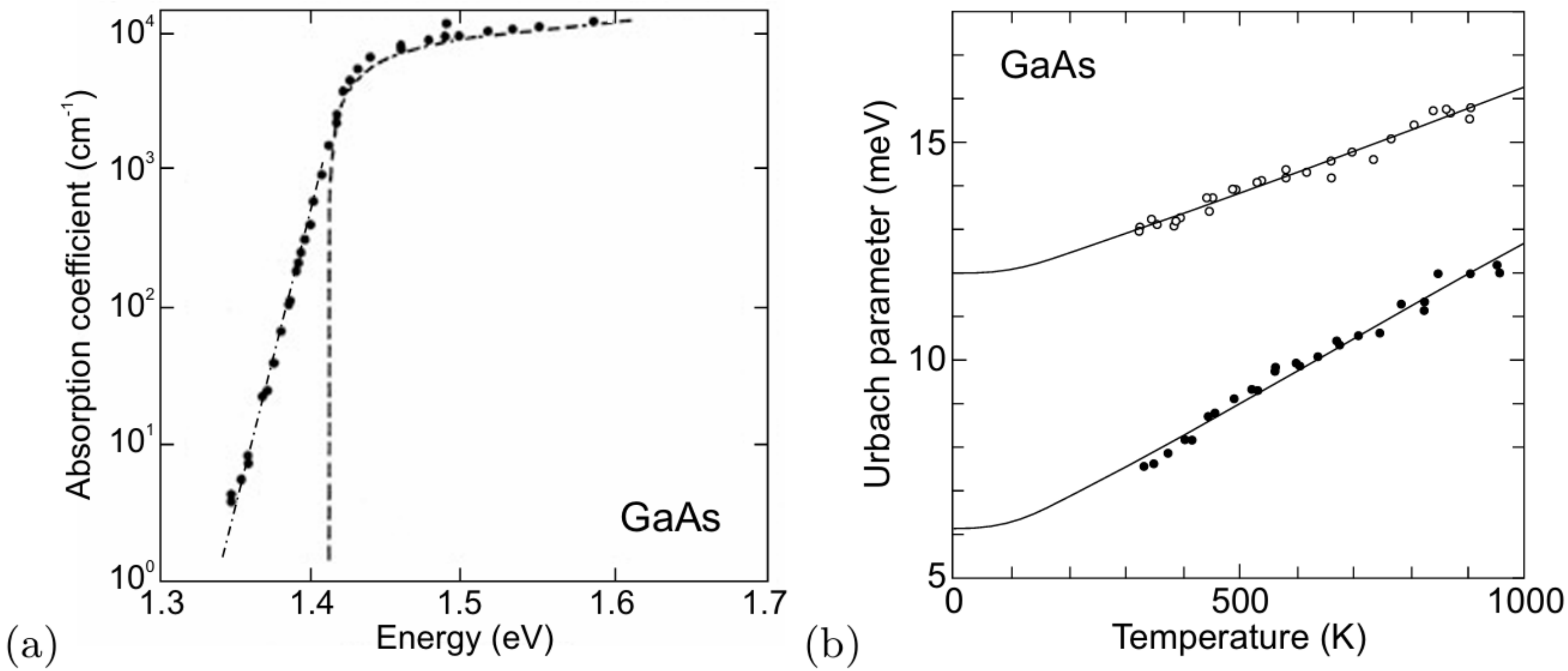
Urbach Tail

- Instead of the ideal $(E - E_g)^{1/2}$ dependence of the direct band-edge absorption, often an exponential tail is observed, This tail is called the Urbach tail, and follows the functional dependence

$$\alpha(E) = \alpha_{\mathrm{g}} \exp\left(\frac{E - E_{\mathrm{g}}}{E_0}\right) ,$$

where E_0 is the characteristic width of the absorption edge, the so-called Urbach parameter

- The Urbach tail is attributed to transitions between band tails below the band edges. Such tails can originate from disorder of the perfect crystal, e.g. from defects or doping, and the fluctuation of electronic energy bands due to lattice vibrations.



Experimental absorption spectrum (circles) of GaAs at room temperature on a semilogarithmic plot. The exponential tail below the bandgap is called the Urbach tail (the dash-dotted line corresponds to E_0 =10.3meV). (b) Temperature dependence of Urbach parameter E_0 for two GaAs samples. Experimental data for undoped (solid circles) and Si-doped ($n = 2\times10^{18}$ cm^{-3}, empty circles) GaAs and theoretical fits (solid lines) with one-phonon model.

13.3 Exciton enhancement of optical absorption

13.3.1 Exciton–Photon Coupling

- Weak coupling: irreversible absorption or recombination
- Strong coupling-Exciton polariton

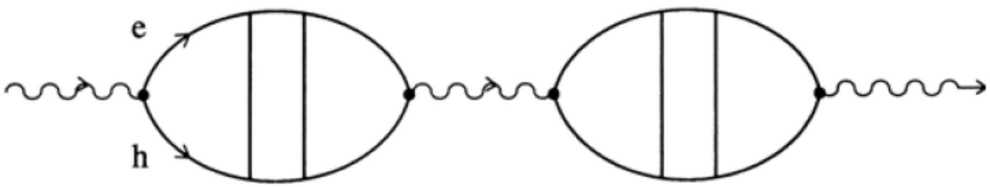
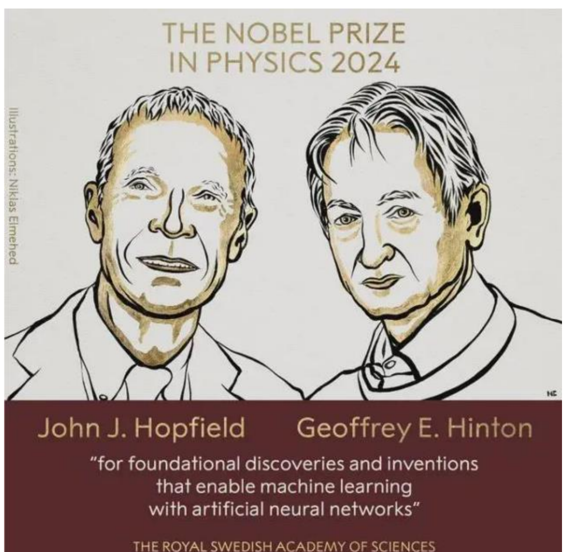


Fig. 13.1. Diagrammatic representation of an exciton polariton



John J. Hopfield, born in Chicago, Illinois, USA in 1933. He received his PhD in Physics from Cornell University in 1958 and currently serves as a professor at Princeton University

➢ The absorption spectra:

- 13.9a an absorption spectrum of ZnO layer at RT
- 13.9b experimental transmission spectra are shown A and $B\Gamma_5$ resonance of a thin CdS platelet type sample
- 13.9c shows a calculated transmission spectrum of the $A\Gamma_5$ $n_B = 1$ resonance
- 13.9d overview of the absorption spectrum of a thin GaAs sample.

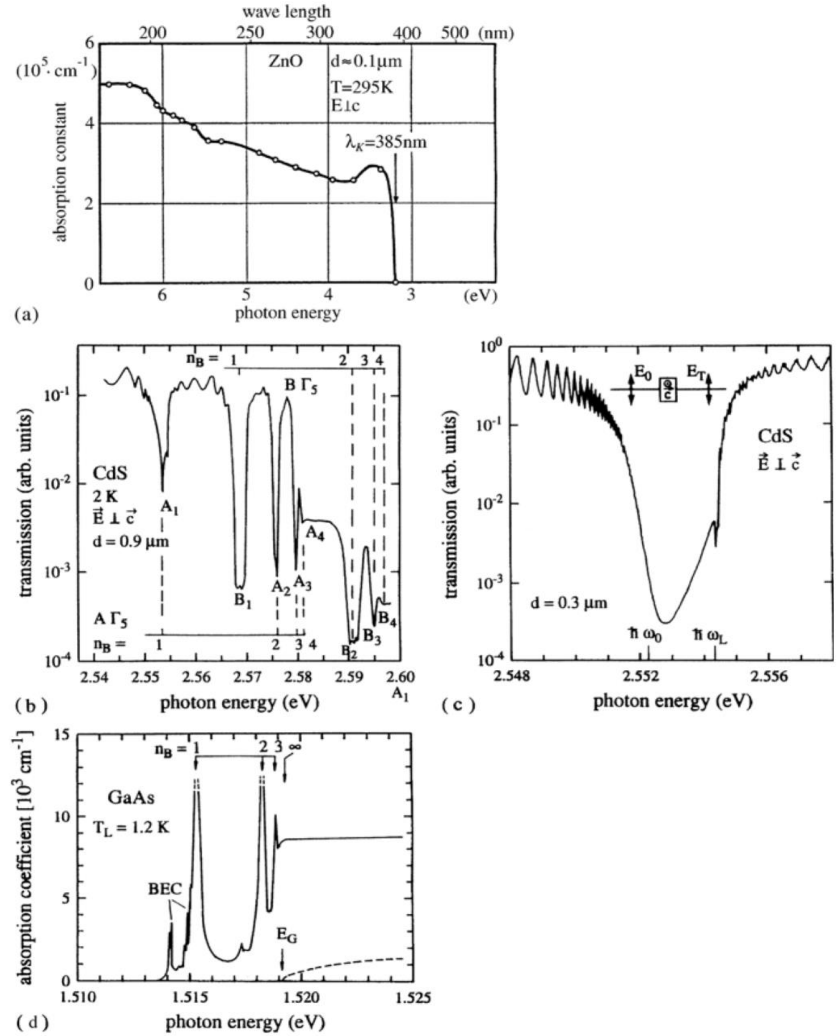


Fig. 13.9. An experimental absorbtion spectrum of ZnO at RT (**a**), an experimental transmission spectrum of CdS for the orientation $\boldsymbol{E} \perp \boldsymbol{c}$ in the region of the A and B exciton resonances (**b**); a calculated one for the $A\Gamma_5$ resonance for $n_B = 1$ (**c**); and an absorption spectrum for a thin GaAs sample (**d**). According to [43M1] to [79V1, 82R1] and to [91U1], respectively

- 13.3.2 Excitons in Organic Semiconductors and in Insulators
- Roughly 600 inorganic semiconductors but only a dozen or so organic ones .
- Hoped that applications as cheap, large area luminescence devices and as electronic devices will be developed.
- Crystalline organic semiconductors usually contain several molecules per unit cell , e.g., anthracene.
- The spectrum of lattice vibrations is complex, with two types of modes: acoustic and optic modes and vibrons or intramolecular modes.
- Concerning the electrons, there are valence bands and conduction bands with a gap between them.
- The bands are rather narrow, resulting in large effective masses of the charge carriers and low mobilities (1~10 cm^2/Vs)
- The interband excitations form excitons with some characteristic difference from the (Wannier) excitons in most inorganic semiconductors.
- Deals in the wide gap ones with Frenkel excitons, electrons and holes reside in the same molecule, resulting in single-triplet splittings.

  The triplet sates: low oscillator strength, low luminescence yield and long lifetime(ms).

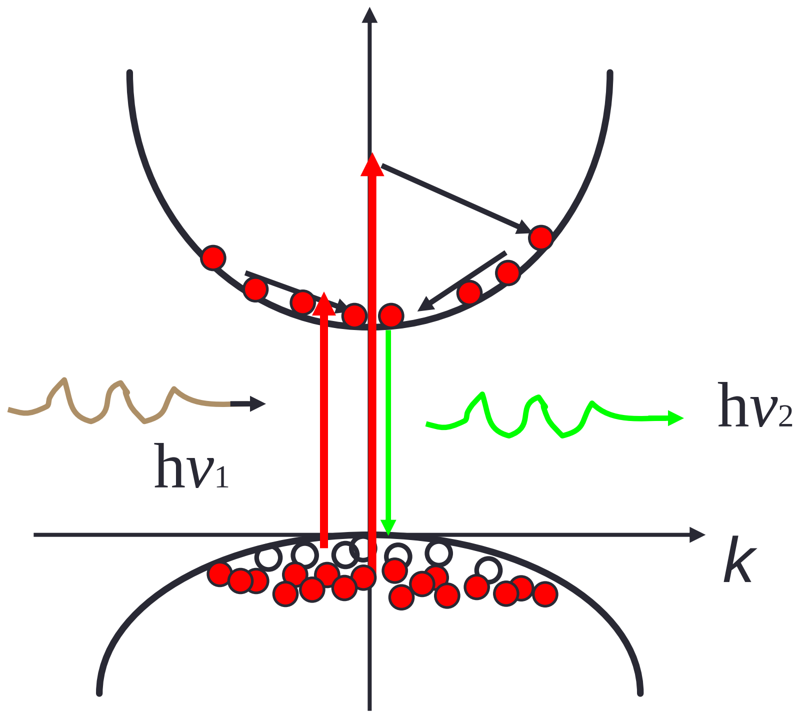
  The single sates: high oscillator strength, and short lifetime(ns).
- Coupling of excitons to phonons may be strong, possibly leading to self-trapping.

13.4 Recombination process

13.4.1 Introduction

- In semiconductors, various excitations (e.g., photon or electron irradiation) may lead to the generation of charge carriers in excess of the thermal equilibrium densities.
- Recombination of electron–hole pairs restores that equilibrium. Recombination processes are distinguished as radiative or nonradiative, depending on whether the recombination results in the emission of a photon or not, respectively
- Luminescence is generally described in terms of radiative recombination of electron–hole pairs, which may involve transitions between states in the conduction or valence bands and those in the energy gap due to donors and acceptors
- Depending on the source of excitation, the luminescence process can be categorized as photoluminescence (photon excitation), electroluminescence (excitation by application of an electric field), and cathodoluminescence (excitation by cathode rays, or energetic electrons).
- In direct-gap semiconductors (e.g., GaAs and CdS), the most likely (direct) transitions are across the minimum energy gap, between the most probably filled states at the minimum of the conduction band and the states most likely to be unoccupied at the maximum of the valence band. Radiative recombination between electrons and holes is relatively likely in such transitions
- In indirect-gap semiconductors (e.g., Si and GaP), since the recombination of electron–hole pairs requires a participation of an extra particle (i.e., phonon), the probability of such a process is significantly lower as compared with direct transitions. Thus, fundamental emission in indirect-gap semiconductors is relatively weak, especially when compared with that due to impurities or defects.

Carrier Relaxation: Four regimes



Ultrashort laser pulse excitation

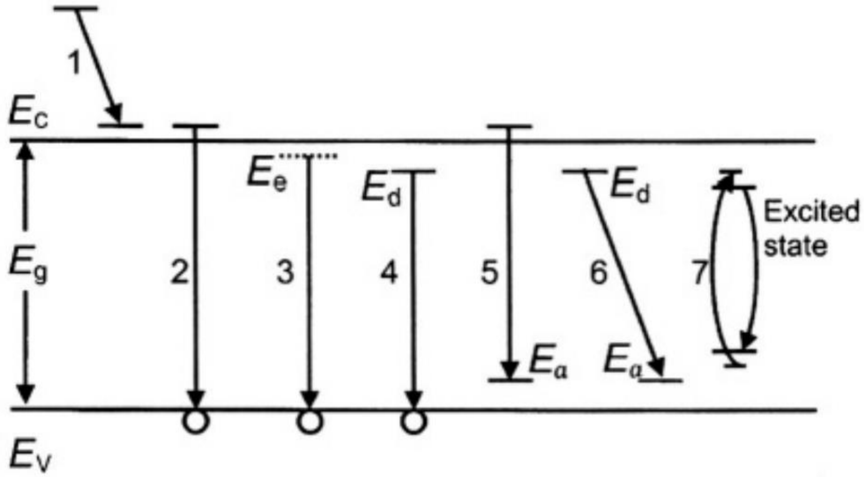
- **Coherent regime (≤200fs)**
  - **Momentum scattering**
  - **Carrier-carrier scattering**
  - **Intervalley scattering (Γ→L,X)**
  - **Hole-optical phonon scattering**
- **Non-thermal regime (≤2ps)**
  - **Electron-hole scattering**
  - **Electron-optical phonon scattering**
  - **Intervalley scattering (L, X→Γ)**
  - **Carrier capture in quantum wells**
  - **Intersubband scattering ($\Delta E>\hbar\omega_{LO}$)**
- **Hot-excitation regime (~1-100ps)**
  - **Hot-carrier-phonon interactions**
  - **Decay of optical phonons**
  - **Carrier-acoustic-phonon scattering**
  - **Intersubband scattering ($\Delta E<\hbar\omega_{LO}$)**
- **Isothermal regime (≥100ps)**
  - **Carrier-recombination**

Time

J. Shah, Ultrafast Spectroscopy of Semiconductors and Semiconductor Nanostructures (Springer, 1998)

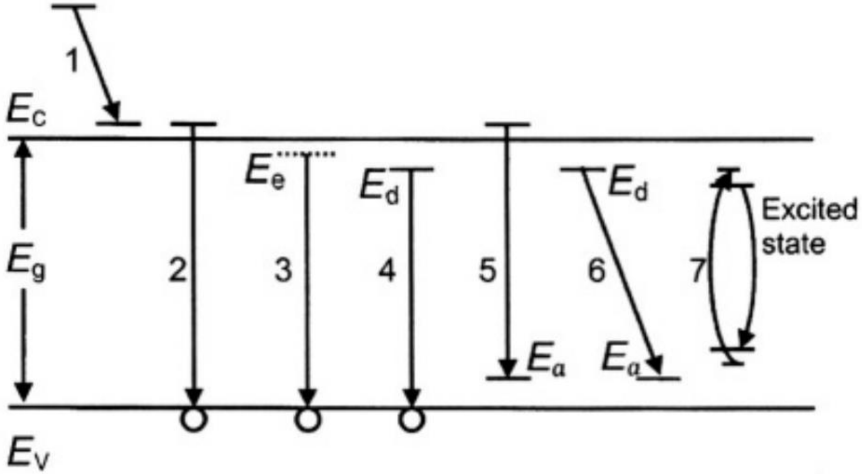
13.4.2 Radiative Transitions

A simplified schematic diagram of radiative recombinations in semiconductors containing impurities is presented in the following figure.



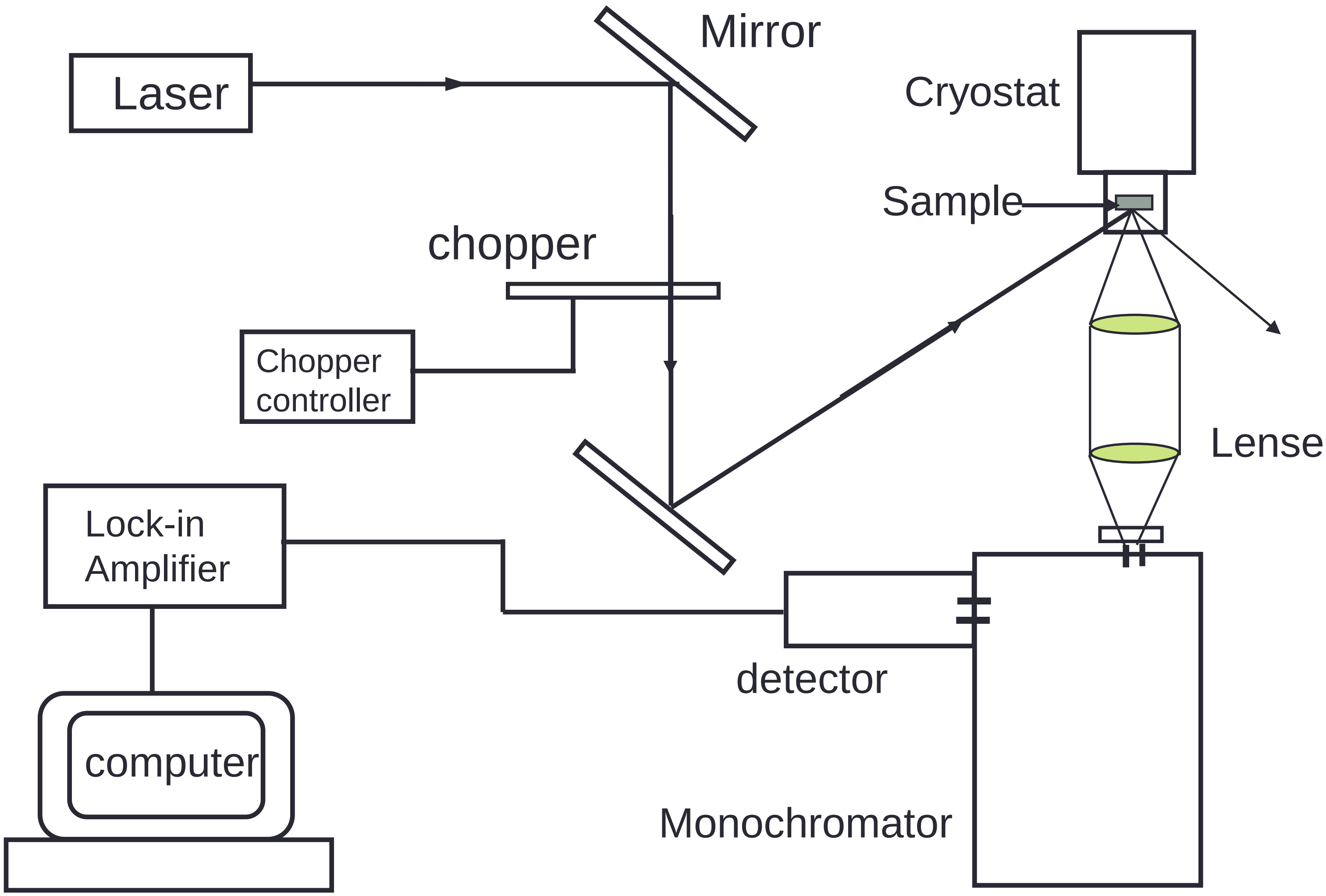
Schematic diagram of radiative recombinations between the conduction band E_c, the valence band E_v, and exciton E_e, donor E_d, acceptor E_a levels in a semiconductor.

- In process 1, which is intraband transition, an electron excited well above the conduction-band edge dribbles down and reaches thermal equilibrium with the lattice. This thermalization process may lead to phonon-assisted photon emission or, more likely, phonon emission only.
- In process 2, which is interband transition, direct recombination between an electron in the conduction band and a hole in the valence band results in the emission of a photon of energy $\hbar\omega \approx E_g$, this transition produces intrinsic luminescence. Although this recombination occurs from states close to the corresponding band edges, the thermal distribution of carriers in these states will lead, in general, to a broad emission spectrum.



- Process 3 is the exciton decay. Both free excitons and excitons bound to an impurity may undergo such transitions. These transitions are often denoted with the following symbols: free-exciton recombination is denoted by X, recombination of an exciton bound at a neutral donor is D^0X, at a neutral acceptor is A^0X, and of excitons bound to the corresponding ionized impurities are D^+X and $A^{-1}X$.
- Processes 4–6 arise from transitions that start and/or finish on localized states of impurities (e.g., donors and acceptors) in the gap.
  - Process 4 represents donor-to-free-hole transition labeled D^0h
  - Process 5 represents free-electron-to-acceptor transition labeled eA^0
  - Process 6 represents the donor–acceptor pair (DAP) recombination model
  - These processes (i.e., 4–6) account for most of the processes in a wide variety of luminescent semiconductor materials. Similar transitions via deep donor and acceptor levels can also occur
- Transition 7 represents the excitation and radiative de-excitation of an impurity with incomplete inner shells, such as a rare earth ion or a transition metal.

13.4.3 Photoluminescence spectroscopy

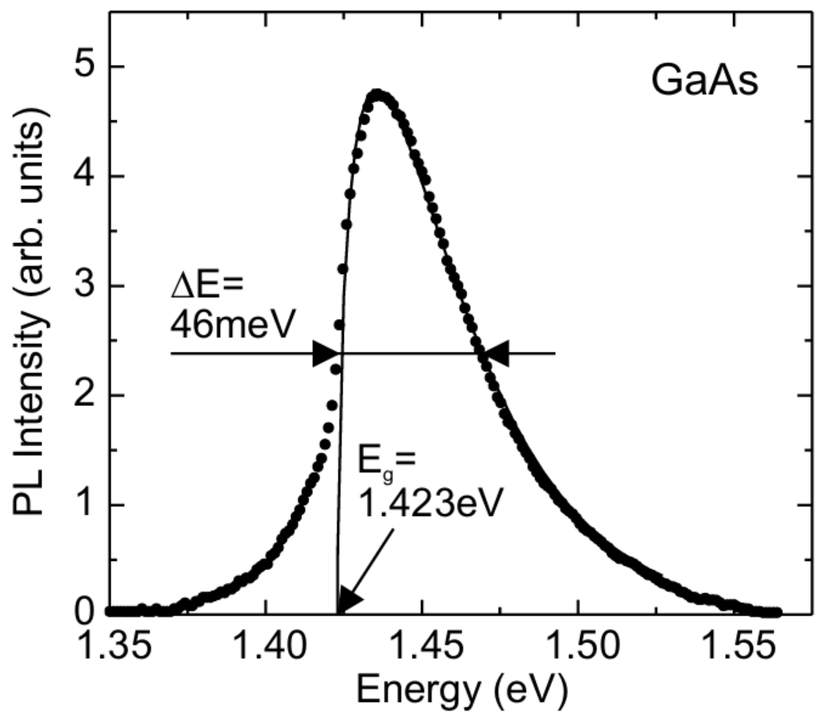


- The spontaneous recombination rate

$$
\begin{aligned}
r_{\mathrm{sp}}(E) &= \int_{E_\mathrm{C}}^{\infty} \mathrm{d}E_\mathrm{e} \int_{-\infty}^{E_\mathrm{V}} \mathrm{d}E_\mathrm{h}\, C(E_\mathrm{e}, E_\mathrm{h}) \\
&\quad \times D_\mathrm{e}(E_\mathrm{e}) f_\mathrm{e}(E_\mathrm{e}) D_\mathrm{h}(E_\mathrm{h}) f_\mathrm{h}(E_\mathrm{h}) \delta(E - E_\mathrm{e} + E_\mathrm{h}) \\
&= \int_{E_\mathrm{C}}^{E+E_\mathrm{V}} \mathrm{d}E_\mathrm{e}\, C(E_\mathrm{e}, E_\mathrm{e} - E) \\
&\quad \times D_\mathrm{e}(E_\mathrm{e}) f_\mathrm{e}(E_\mathrm{e}) D_\mathrm{h}(E_\mathrm{e} - E) f_\mathrm{h}(E_\mathrm{e} - E)\ ,
\end{aligned}
$$

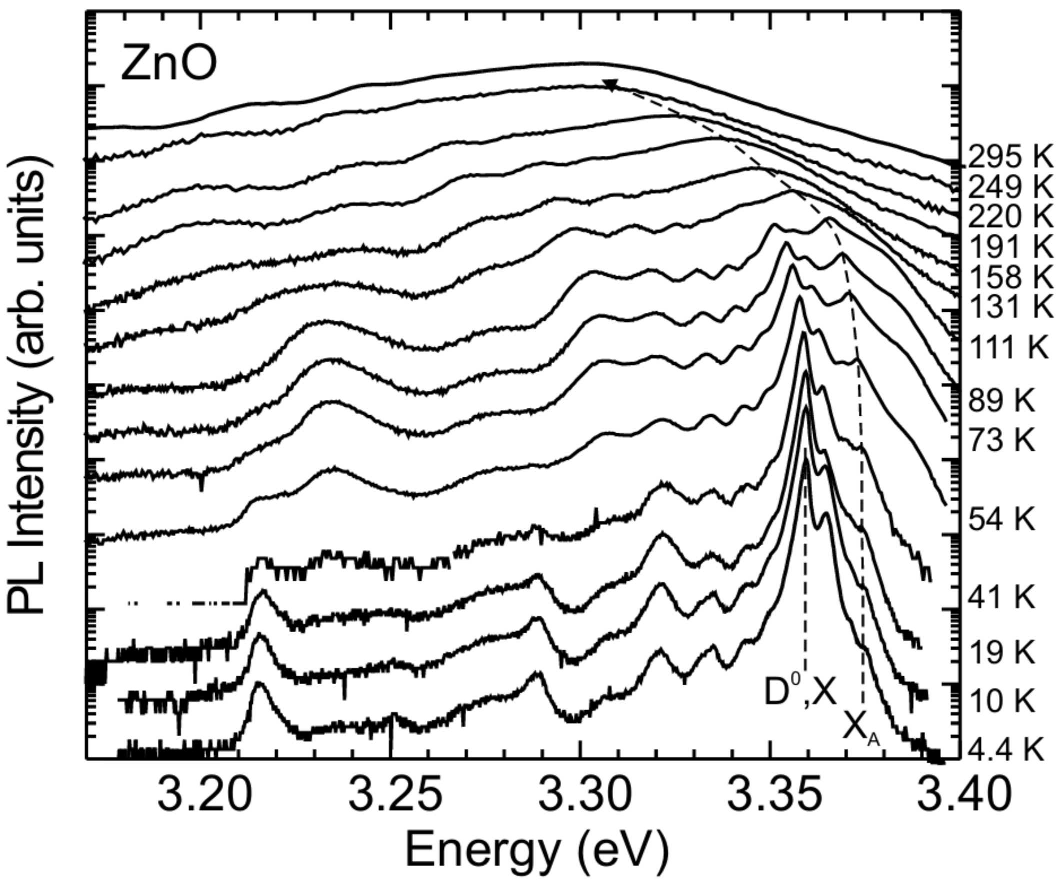
At small excitation and at low doping it can be approximated by the Boltzmann distribution function and the lineshape is given as

$$
I(E) \propto (E - E_\mathrm{g})^{1/2} \exp\left(-\frac{E}{kT}\right) . \qquad (13.4.1)
$$



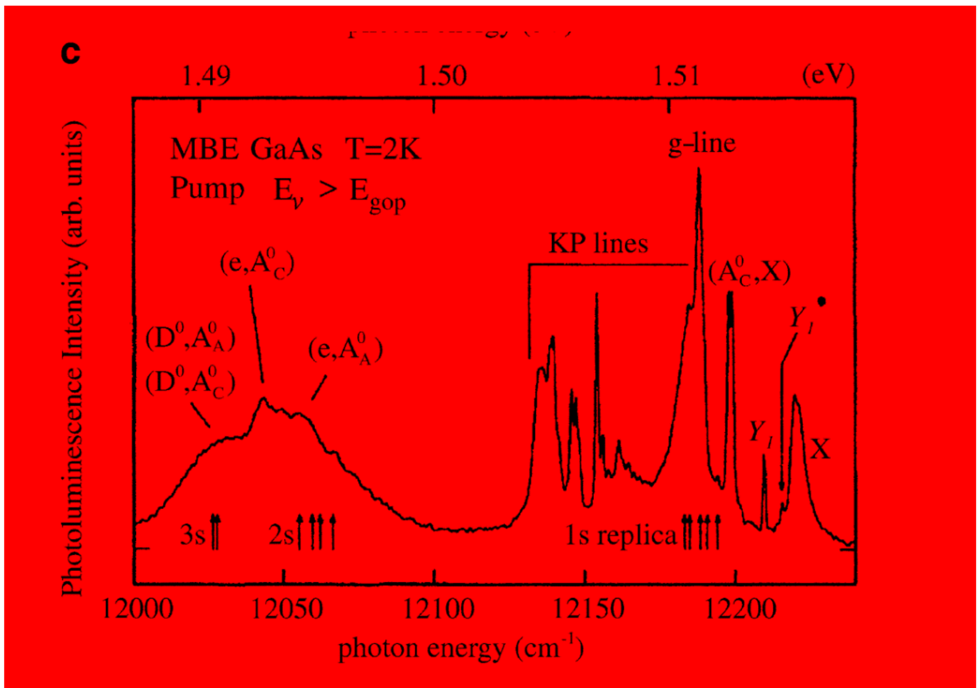
Photoluminescence spectrum of an undoped LPE-grown epitaxial GaAs layer at room temperature and low cw (λ = 647nm) excitation density (10W/cm^2). The solid line is a lineshape fit with (13.4.1).

Free exciton recombination



Temperature-dependent luminescence spectra of a ZnO thin film (on sapphire). At low temperatures, the spectra are dominated by donor-bound exciton transitions. The vertical dashed line indicates the low-temperature position of the donor-bound exciton transition (D^0,X). The curved dashed line visualizes the energy position of the free-exciton transition (X_A) that becomes dominant at room temperature.

- BEC often show up in luminescence and absorption spectra as extremely narrow peaks.
- Chemical shift or central cell correction：
  a splitting of the lines caused by the chemical nature of the binding atom.
- the low temperature luminescence of high quality samples is generally dominated by defect (or localiation-) related recombination processes.



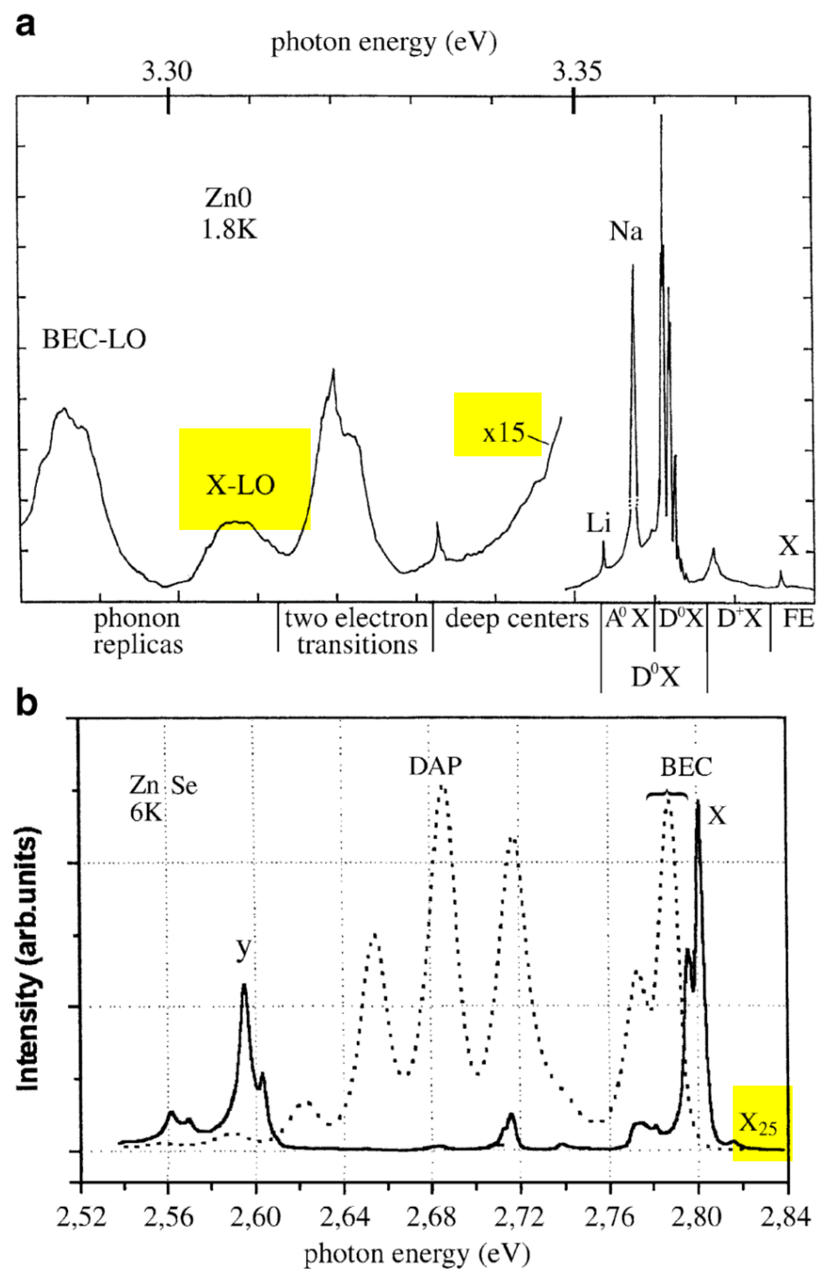


Fig. 14.2 Low-temperature and low-excitation luminescence spectra of ZnO (a) of two differently grown ZnSe samples (solid line: MBE growth with element sources, dashed line: with a compound source) (b) and of GaAs (c)

- Recombination process : free-to-bound transition

The corresponding emission peak is at :

$$\hbar\omega_{FB} = E_g - E^b_{D^0/A^0} - m\hbar\omega_{LO} \qquad (13.4.2)$$

- Recombination process with DA pairs

The corresponding emission peak is at :

$$\hbar\omega_{FB} = E_g - E^b_{D^0} - E^b_{A^0} - m\hbar\omega_{LO} \qquad (13.4.3)$$

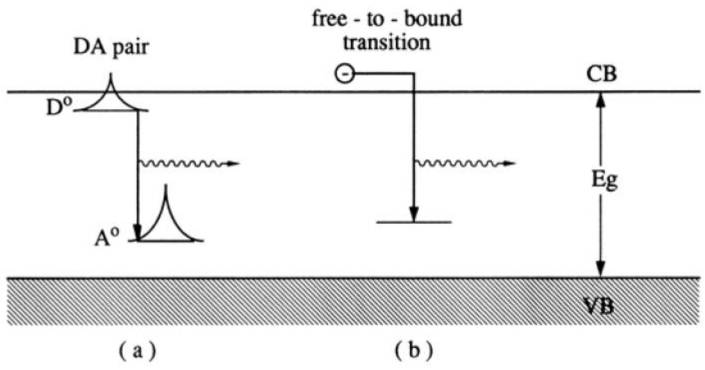


Fig. 14.9. A donor–acceptor pair (**a**) and a free-to-bound recombination process (b

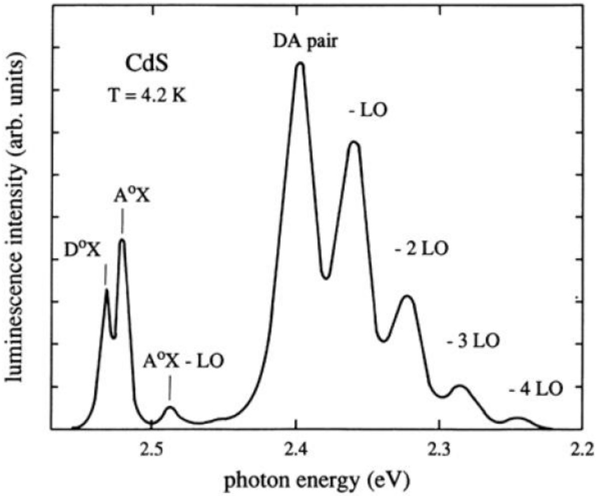


Fig. 14.10. Donor–acceptor pair luminescence in CdS including some phonon replica [79B1]

Indirection transitions in indirector semiconductors

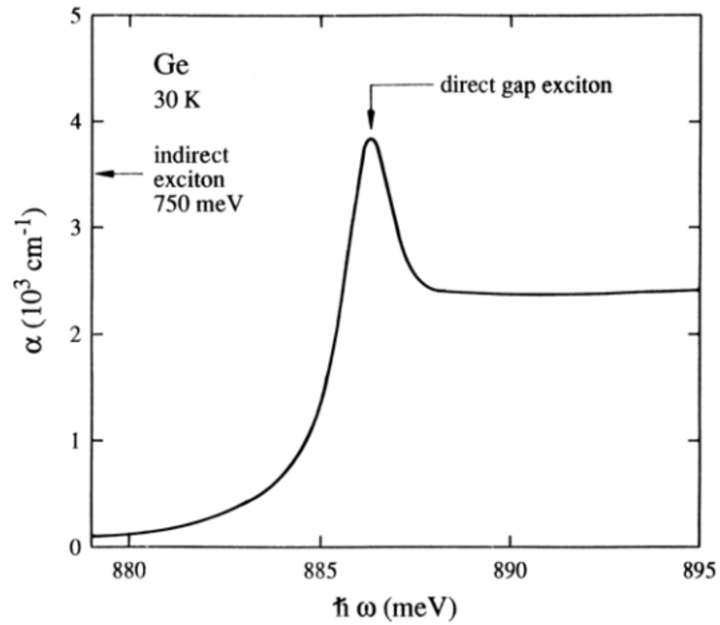


Fig. 13.34. Absorption around the direct exciton in Ge. According to [83S2]

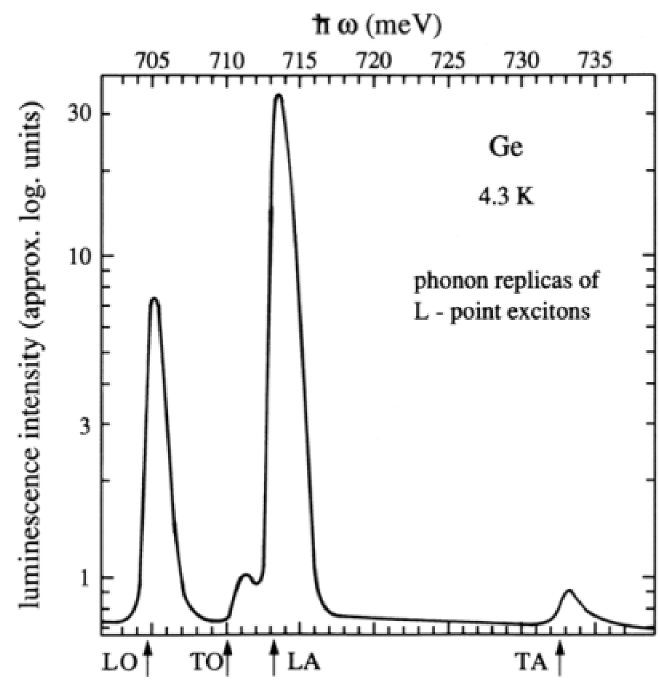


Fig. 13.35. A luminescence spectrum of Ge showing various phonon satellites. According to [76T1]

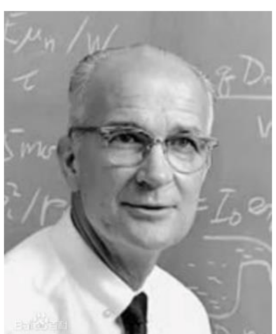
13.4.3 Nonradiative recombinations

The electron–hole pair recombination may occur via nonradiative processes as well. Nonradiative recombination processes include：

- multiple-phonon emission - direct conversion of the energy of an electron to heat
- Auger effect- the energy of an electron transition is absorbed by another electron which is raised to a higher-energy state in the conduction band, with subsequent emission of the electron from the material or dissipation of its energy through emission of phonons
- recombination due to surface states and defects

The kinetics of the recombination processes, related to the minority carrier capture at defects having levels in the energy gap of the semiconductor, can be described by the Shockley–Read–Hall (SRH) recombination model. According to this model, for a single trapping energy level present in the energy gap, the SRH recombination rate can be expressed as

$$R_{\mathrm{SRH}} = \frac{N_t \sigma_n \sigma_p v_{th} (np - n_i^2)}{\sigma_n \{n + n_i \exp[(E_t - E_i)/k_B T]\} + \sigma_p \{p + n_i \exp[(E_i - E_t)/k_B T]\}} \tag{13.4.4}$$



- 因在贝尔实验室期间与同事共同发明晶体管，获得1956年诺贝尔物理奖。晶体管在很大程度上取代了体积更大、效率更低的真空管，开创了微型电子时代，他还获得了90多项发明专利
- 1955年，他在加州芒创立了「肖克利实验室股份有限公司」，聘用了很多年轻优秀的人才。但很快肖克利个人的管理方法因其公司内部不合，八名主要员工（八叛逆）于1957年集体跳槽成立了仙童半导体公司，仙童半导体发明了集成电路

Further discussion on the SRH recombination rate quation

$$R_{\mathrm{SRH}} = \frac{N_t \sigma_n \sigma_p v_{th}(np - n_i^2)}{\sigma_n\{n + n_i \exp[(E_t - E_i)/k_B T]\} + \sigma_p\{p + n_i \exp[(E_i - E_t)/k_B T]\}}$$

σ_n and σ_p are theelectron and hole capture cross-sections, respectively
N_t is the volume density of deep levels
V_{th} is the carrier thermal velocity
E_t is the energy level of the trap
R has units of $cm^{-3}s^{-1}$
The term np-n_i^2 in this equation indicates the deviation from the thermal equilibrium conditions, and for np=n_i^2 (i.e., for the thermal equilibrium condition) the recombination rate is 0

13.4.4 Recombination rate

Various excitations may lead to the generation of charge carriers in excess of the thermal equilibrium densities, and the recombination of electron–hole pairs restores that equilibrium. The recombination rate R is proportional to the product of the concentration of occupied states (i.e., electrons) in the conduction band and the concentration of unoccupied states (i.e., holes) in the valence band. $R=Bn_0p_0$,

where B is a constant and n_0 and p_0 are the equilibrium concentrations of electrons and holes. If Δn and Δp are the excess carrier concentrations, we can write

$$-\frac{d\Delta n(t)}{dt} = B[n_0 + \Delta n(t)][p_0 + \Delta p(t)] - Bn_i^2 \qquad (13.4.5)$$

For a p-type semiconductor, this equation can be simplified to

$$-\frac{d\Delta n(t)}{dt} = Bp_0\,\Delta n(t) \qquad (13.4.6)$$

Thus, if the generation process ceases, the initial excess carrier concentration will decay exponentially

$$\Delta n(t) = \Delta n(0)\exp(-Bp_0 t) = \Delta n(0)\exp(-t/\tau_n) \qquad (13.4.7)$$

Where $\tau_n=(Bp_0)^{-1}$ is called the recombination lifetime or minority carrier lifetime. Similarly, the decay of excess holes in n-type semiconductor occurs with decay constant, $\tau_p=(Bn_0)^{-1}$

- In general, the description of such optical processes as the formation of the luminescence involves the analysis of the generation, diffusion, and recombination of minority carriers. Under equilibrium conditions, the generation of electron–hole pairs is balanced by recombination processes. These recombination processes may include radiative band-to-band and band-to-impurity recombination, and nonradiative recombination processes such as trap-assisted recombination, Auger recombination, and surface recombination.
- For both competitive radiative and nonradiative centers present, the observable lifetime is

$$\frac{1}{\tau}=\frac{1}{\tau_r}+\frac{1}{\tau_{nr}} \qquad \tau=\frac{\tau_r\tau_{nr}}{\tau_r+\tau_{nr}} \tag{13.4.8}$$

- In general, τ_{nr} is the resultant of several nonradiative recombination processes that may be present in a given material,

$$\frac{1}{\tau_{nr}}=\sum_i\frac{1}{\tau_{nri}} \tag{13.4.9}$$

- The radiative recombination efficiency (or internal quantum efficiency) η which is defined as the ratio of the radiative recombination rate R_r to the total recombination rate R, can be written as

$$\eta=\frac{\tau}{\tau_r}=\frac{1}{1+(\tau_r/\tau_{nr})} \tag{13.4.10}$$

- When the nonradiative recombination process is faster, radiative recombination efficiency is relatively small. On the other hand, when the radiative recombination process is faster, radiative recombination efficiency is relatively greater

- For a semiconductor that contains only one type of radiative and one of nonradiative recombination center, the radiative recombination efficiency can be expressed as, using a relationship $\tau = (N\sigma v_{th})^{-1}$

$$\eta = \frac{\tau}{\tau_r} = \frac{1}{1 + (N_{nr}\sigma_{nr}/N_r\sigma_r)} \qquad (13.4.11)$$

  where N_r and N_{nr} are the densities of the radiative and nonradiative recombination centers, respectively, σ_r and σ_{nr} are the radiative and nonradiative capture cross sections, and v_{th} is the carrier thermal velocity.

- In general, η depends on temperature, the particular recombination centers and their concentrations, and the presence of various defects.

- In principle, it is possible in some cases to ensure that $N_r > N_{nr}$, but typically, $\sigma_{nr} >> \sigma_r$

- In semiconductors, the measured minority carrier lifetime depends on various recombination mechanisms.

- For bulk semiconductors (including thick semiconductor layers), the minority carrier lifetime

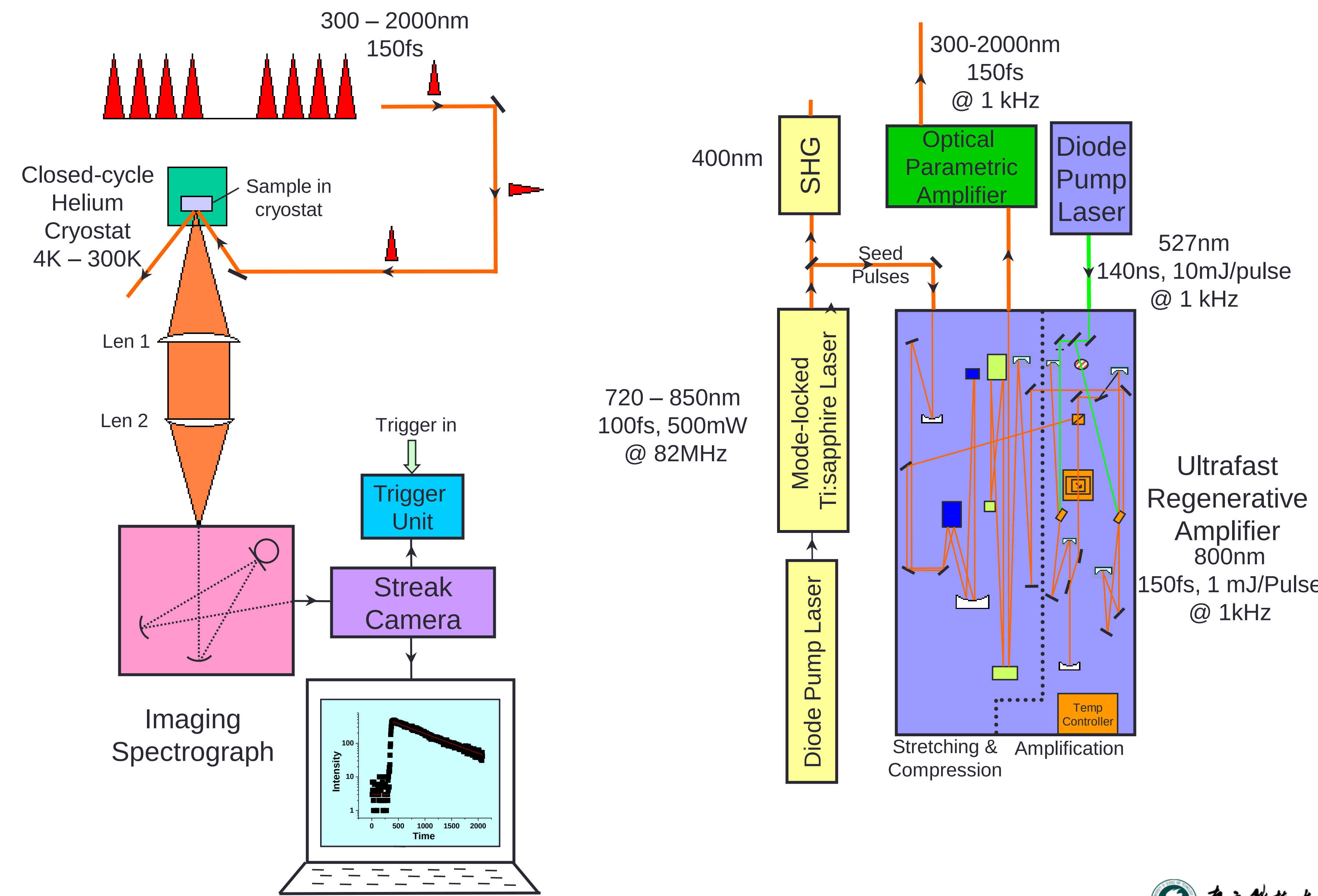
$$\frac{1}{\tau_{bulk}} = \frac{1}{\tau_r} + \frac{1}{\tau_{nr}} \tag{13.4.12}$$

- In low-dimensional semiconductors, one has also to include additional recombination mechanism, i.e., nonradiative surface (or interface) recombination. Thus, the measured values of the lifetime are effective values determined by the bulk and surface lifetimes

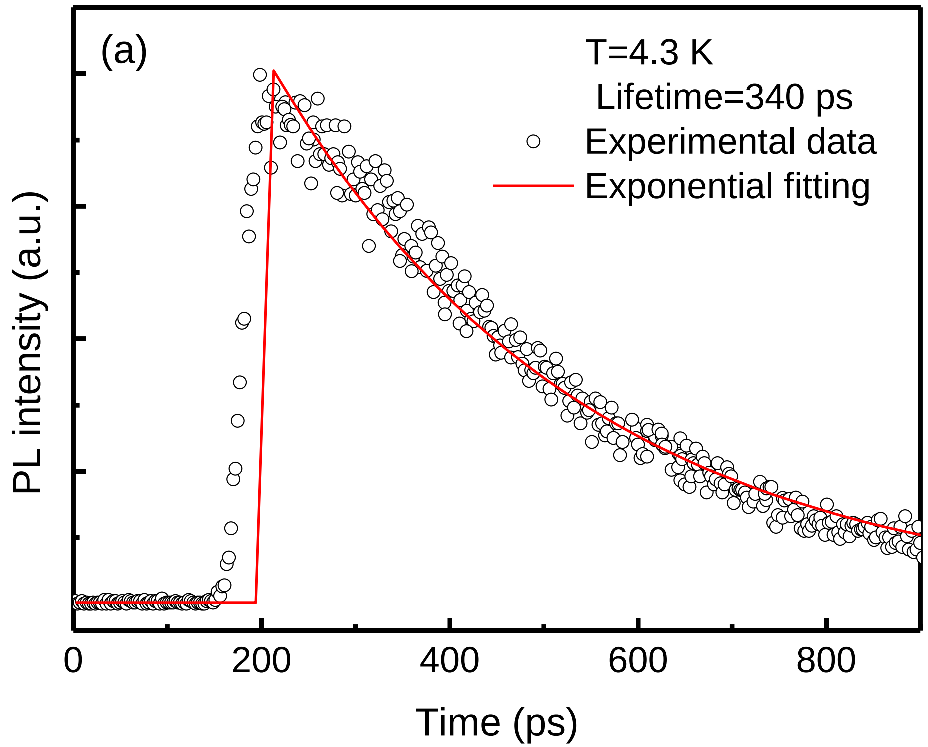
$$\frac{1}{\tau_{measured}} = \frac{1}{\tau_r} + \frac{1}{\tau_{nr}} + \frac{1}{\tau_{surf}} \tag{13.4.13}$$

- In general, whereas τ_{bulk} depends on the density of recombination centers in the bulk of the material, τ_{surf} is determined to a large extent by the recombination centers on the surface.
- Surface states arise due to the abrupt change at the surface of the three-dimensional band structure associated with the bulk of the material.
- In addition, impurity atoms and oxide layers (on the surface of the material) may also produce discrete energy levels.
- In pure crystals, τ_{bulk} is typically long, so that surface recombination will dominate. In such a case, the lifetime of the material may depend strongly on surface treatment procedures.

Experimental Setup – Time-Resolved PL



PL decay of ZnO



$$I_{PL}=I_0\exp(t/\tau)$$

Questions